NetGuard on a New Raspberry Pi — Installation Guide

Version: 1.0
Repository: https://github.com/jibinmichael89-spec/netguard

Overview

NetGuard is a home network security monitor for Raspberry Pi. It discovers devices on your LAN, scans open ports, scores risk, monitors DNS, and provides a web dashboard.

After installation:

- **Dashboard URL:** `http://<pi-ip-address>:8000`
- **Auto-start:** All core services start on boot via systemd
- **Install location:** `/opt/netguard`
- **Database:** `/var/lib/netguard/netguard.db`

Services installed automatically

Service	Purpose	Runs as
netguard-api	Web dashboard + REST API (port 8000)	netguard
netguard-arp-scanner	Device discovery + port scanning	root
netguard-risk-scorer	Composite risk scoring	netguard
netguard-dns-monitor	DNS query capture	root

Optional (disabled by default):

Service	Purpose
netguard-network-blocker	ARP isolation for blocked devices (limited on mesh WiFi)

Requirements

Item	Details
Hardware	Raspberry Pi 3, 4, or 5 with power supply
Operating system	Raspberry Pi OS (64-bit recommended)
Network	Pi connected to your home LAN (Ethernet recommended)
Access	SSH enabled, or keyboard + monitor
Internet	Required during install (apt, pip, npm)

Part 1 — Prepare the Raspberry Pi (One-Time Setup)

Step 1: Flash Raspberry Pi OS

1. Download **Raspberry Pi Imager** from https://www.raspberrypi.com/software/
2. Flash **Raspberry Pi OS (64-bit)** to a microSD card
3. In Imager settings (gear icon), enable:
4. **SSH** (password or key authentication)
5. **Username and password** (example: user `netguard`)
6. **WiFi** (if not using Ethernet)
7. Insert the card, power on the Pi, and wait for it to boot

Step 2: Find the Pi's IP address

From your PC:

```
ping raspberrypi.local
```

Or check your router's connected-devices list. Example IP: `192.168.1.71`

Step 3: Connect via SSH

```
ssh netguard@192.168.1.71
```

Replace `netguard` with your username and use your Pi's IP address.

Step 4: Update the operating system

```
sudo apt update
sudo apt upgrade -y
sudo reboot
```

SSH back in after the reboot completes.

Part 2 — Install NetGuard

Step 1: Install Git

```
sudo apt install -y git
```

Step 2: Clone the repository

```
cd ~
git clone https://github.com/jibinmichael89-spec/netguard.git
cd netguard
```

Step 3: Run the installer

```
chmod +x install/pi/install.sh install/pi/uninstall.sh
sudo install/pi/install.sh
```

The installer automatically:

1. Installs system packages (Python, Scapy, libpcap, iptables)
2. Creates the `netguard` system user if needed
3. Copies the application to `/opt/netguard`
4. Creates a Python virtual environment and installs dependencies
5. Builds the web dashboard (installs Node.js/npm if needed)
6. Creates data directories and configuration files
7. Installs and enables systemd services
8. Starts all core services immediately

Expected install time: 5–15 minutes (dashboard npm build is the slowest step).

Step 4: Open the dashboard

When installation completes, the script prints your dashboard URL:

```
Dashboard: http://192.168.1.71:8000
```

Open that address in a browser on any device on your home network.

Part 3 — Verify Installation

Run these commands on the Pi:

```
# Check all services
sudo systemctl status netguard.target

# Test API health endpoint
curl -s http://127.0.0.1:8000/health

# List discovered devices (may take 30–60 seconds on first run)
curl -s http://127.0.0.1:8000/devices
```

Success indicators:

- `netguard.target` shows all services as **active (running)**
- `/health` returns a successful response
- Devices appear in the dashboard within about one minute

Part 4 — Day-to-Day Management

Restart all NetGuard services

```
sudo systemctl restart netguard.target
```

View live logs

```
sudo journalctl -u netguard-api -f
sudo journalctl -u netguard-arp-scanner -f
```

Check service status

```
sudo systemctl status netguard-api
sudo systemctl status netguard-arp-scanner
sudo systemctl status netguard-risk-scorer
sudo systemctl status netguard-dns-monitor
```

Optional Configuration

DNS logging via dnsmasq

If your Pi runs dnsmasq as the LAN DNS server, add to `/etc/dnsmasq.conf`:

```
log-queries
log-facility=/var/log/netguard/dnsmasq.log
```

Then restart dnsmasq:

```
sudo systemctl restart dnsmasq
```

Enable network block enforcer

```
sudo systemctl enable --now netguard-network-blocker.service
```

Note: On mesh WiFi systems (Linksys Velop, Eero, Orbi), ARP-based blocking is often ineffective. Use your router's parental controls or device-pause feature for guaranteed enforcement.

Alternative Install Method — Offline Tarball

Use this when the Pi cannot clone from GitHub directly.

On your development PC (with the repository):

```
cd netguard
chmod +x install/pi/build-release.sh
./install/pi/build-release.sh
scp dist/NetGuard-pi-*.tar.gz netguard@<pi-ip>:~/
```

On the Raspberry Pi:

```
tar xzf NetGuard-pi-*.tar.gz
cd NetGuard-pi
chmod +x install.sh uninstall.sh
sudo ./install.sh
```

Uninstall NetGuard

```
cd ~/netguard
sudo install/pi/uninstall.sh
```

To keep the database and logs:

```
sudo install/pi/uninstall.sh --keep-data
```

File Locations Reference

Item	Path
Application	`/opt/netguard`
Database	`/var/lib/netguard/netguard.db`
Logs	`/var/log/netguard/`
Environment config	`/etc/netguard/netguard.env`
Systemd units	`/etc/systemd/system/netguard-*.service`

Troubleshooting

Problem	Solution
Dashboard won't load	Run `sudo systemctl status netguard-api` and check logs with `sudo journalctl -u netguard-api -n 50`
No devices in dashboard	Ensure `netguard-arp-scanner` is running; wait 60 seconds after boot
`install/pi/install.sh` not found	Run `git pull` in the repository to get the latest files
Port 8000 unreachable from other PCs	Check firewall: `sudo ufw allow 8000` (if ufw is enabled)
Block button doesn't stop internet	Expected on mesh WiFi; use router app to pause the device

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Quick Install Cheat Sheet

```
sudo apt update && sudo apt install -y git
git clone https://github.com/jibinmichael89-spec/netguard.git
cd netguard
chmod +x install/pi/install.sh
sudo install/pi/install.sh
```

Then open: **http://\<pi-ip>:8000**

NetGuard — Home network security monitor